

جـامعة بيروت العربية
BEIRUT ARAB UNIVERSITY

Faculty of Science

Department of Mathematics and Computer Science

Deep Learning (CMPS470)

Nonlinear House Price Modeling with Deep Neural Networks

Student Name: **Student ID:**

Nour Ajami 202302485

Supervisor:

Dr. Mahmoud Skafi Supervisor

T.A. Lynn El Arwadi Lab Instructor

Introduction

House price prediction is a difficult machine learning problem. House prices are a nonlinear function of many different parameters including the size of the house, the quality of the neighborhood, age of the house and general condition. The Ames Housing dataset is a collection of characteristics for over 1500 homes in Ames, Iowa with corresponding prices. The data is mixed with both numerical and categorical features and contains missing values making this an even more difficult problem.

The main goal of this project is to compare between using a shallow neural network and deep neural network to estimate house prices. I also investigated how deep learning models learn hidden features from data.

Data Preprocessing and Feature Engineering

I tried out different things for preprocessing before going on to training the model. For the numerical missing values, I filled in the median and for the categorical missing values I just changed the NAs to “None”.

To make the input understandable for the neural network, I applied one hot encoding on two categorical variables house types and neighborhood.

New features were added to improve model learning:

- TotalArea = TotalBsmtSF + 1stFlrSF + 2ndFlrSF
- HouseAge = YrSold – YearBuiltt
- GarageAge = YrSold – GarageYrBlt

The target variable was set to be logarithmic transformation of SalePrice, which was right skewed. Three more features were also skewed to the right (GrLivArea, LotArea, TotalBsmtSF).

Model Design and Experimental Setup

Four models were trained and compared:

- Linear Regression
- 1-layer Neural Network

- 3-layer Neural Network
- 5-layer Neural Network

The two models were compared by training and testing them under the same conditions. Each model was trained on the same split with the same amount of data augmentation. An 80/20 ratio split was used between training and data.

I also used evaluation metrics like:

- Mean Squared Error (MSE)
- Mean Absolute Percentage Error (MAPE)

Results and Comparison

Instead of utilizing a linear regression approach, I used neural networks to better learn the relationships between the features in the training data. The best result was with the 3-layer network which showed a balance between learning and generalization. The 5-layer network showed lower loss on the training data, but it started to overfit because the loss on the validation set did not improve.

Analysis and Discussion

In this experiment feature engineering played an important role in the results on the deep learning modeling. TotalArea and HouseAge were two of the additional features that helped better understand the properties and gave chance for a better pricing.
As the model increased, the accuracy improved but the model started to overfit around depth of 34 to 36 layers. This is seen in the generalization gap plot for each run.
The log of the price was used in the end as it improved the stability of the fit and reduced the errors for the expensive houses.

Representation Insight

To better understand and get idea of what the neural network was learning, I used a hidden layer activation from best performing network through the PCA.

The PCA visualization showed that there is some clustering and some of the neighborhoods were separated out. This means that the network is learning meaningful hidden representations, and the information about the location of the house and the quality of the house.

Conclusion

The goal of the project is to use deep learning to predict the house price given features of the property. This is done by trying to learn nonlinear relationships between the input features and the target price using a deep learning mode.
Preprocessing, feature engineering, and log transformation all helped. Increasing the depth helped learning up to a certain point at which very deep models overfit.